

Notes: Giroud and Rauh (JPE, 2019)
**State Taxation and the Reallocation of Business Activity: Evidence from
Establishment-Level Data**

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1 Big picture

- **Question.** How do state-level business taxes affect employment, establishments, and capital when firms can reallocate activity across states and choose different federal tax organizational forms?
- **Core empirical idea.** The paper exploits the fact that state corporate taxes directly apply to C corporations, while state personal taxes directly affect pass-through entities such as S corporations, partnerships, and sole proprietorships.
- **Main data advantage.** The paper uses confidential Census establishment-level microdata and observes firms operating in multiple states. This allows the authors to track establishment counts, employment, and capital within the same firm across different state tax environments.
- **Main estimate.** For C corporations, employment and establishment counts have short-run corporate-tax elasticities around -0.4 to -0.5 . These outcomes are largely invariant to personal tax rates.
- **Pass-through estimate.** For pass-through entities, employment and establishment counts respond to personal income tax rates, with elasticities around -0.2 to -0.4 , and are largely invariant to corporate tax rates.
- **Capital result.** Manufacturing capital shows the same pattern: C corporation capital responds to corporate taxes, while pass-through capital responds to personal taxes.
- **Reallocation result.** Roughly half of the effect comes from reallocation of productive resources to other states rather than pure contraction of the firm.
- **Heterogeneity.** Responses are strongest in tradable and footloose industries, where firms can more easily serve demand from alternative locations.
- **Takeaway.** State taxation affects the geography of business activity. The effect is not merely an aggregate state-level correlation; it appears within multistate firms and depends on the tax code that applies to the firm's organizational form.

2 Incremental contribution of the paper

2.1 Contribution relative to state tax and economic growth literature

- Earlier work often uses aggregate state, county, or industry panels to estimate the relationship between taxes and employment or growth.
- These aggregate studies face two major limitations: tax policy may respond to economic conditions, and aggregate data cannot tell whether firms move activity across states or reduce activity overall.
- Giroud and Rauh use establishment-level Census data, allowing them to observe the location of business activity within the same firm over time.

Interpretation: The paper moves the literature from aggregate correlations to within-firm, across-state reallocation. This is the central empirical advance.

2.2 Contribution relative to plant-location studies

- Plant-location studies usually focus on entry of new establishments or firms.
- This paper studies both the extensive margin, measured by the number of establishments a firm operates in a state, and the intensive margin, measured by employment and capital at existing establishments.
- It also separates own-state effects from other-state tax effects, thereby measuring reallocation within multistate firms.

Interpretation: The paper shows not just whether plants are located in low-tax states, but whether firms actively shift resources away from high-tax states toward lower-tax alternatives.

2.3 Contribution from organizational-form comparisons

- The paper’s key identification insight is to compare C corporations with pass-through entities.
- C corporations should respond to corporate tax changes, but not personal tax changes.
- Pass-through entities should respond to personal tax changes, but not corporate tax changes.
- The evidence follows this pattern closely, which helps rule out broad omitted state shocks as the main driver.

Interpretation: The cross-organizational-form placebo structure is powerful. A state shock that depresses local business activity should affect both organizational forms, while the paper finds tax-code-specific responses.

2.4 Contribution to corporate finance and public finance

- Corporate finance often studies firms as units of decision making, while public finance often studies tax bases and aggregate responses.
- This paper bridges the two by showing that multistate firms reallocate real activity in response to state taxation.
- The result matters for state tax policy because tax increases can shrink the local tax base and move jobs and capital elsewhere.

Interpretation: The paper’s incremental contribution is both methodological and substantive: it provides micro evidence for real activity reallocation as a central margin of business tax avoidance.

3 Data and measurement

3.1 Census establishment data

- The main establishment-level data come from the Longitudinal Business Database (LBD), covering firms and establishments from 1977 to 2011.
- The analysis focuses on firms with at least 100 employees and establishments in at least two states.
- The LBD provides establishment location, firm ownership, employment, and legal form of organization.
- The manufacturing capital analysis uses the Annual Survey of Manufactures and Census of Manufactures, linked to the LBD.

Interpretation: Multistate firms are essential because they allow comparisons of the same firm’s activity across different tax regimes.

3.2 Legal form and tax treatment

- C corporations pay corporate income taxes at the entity level.
- S corporations, partnerships, and sole proprietorships are pass-through entities: income flows through to owners and is taxed under the individual income tax code.
- LLCs are assigned according to their actual tax filing status.

- The paper uses legal form to predict which tax code should affect firm behavior.

Interpretation: This organizational-form distinction is the empirical source of falsification. If corporate taxes affect pass-through establishments, or if personal taxes affect C corporations, the tax-code interpretation would be weaker.

3.3 Tax variables

- τ_{st}^C is the top state corporate income tax rate.
- τ_{st}^P is the top state personal income tax rate.
- Additional controls include property tax share, unemployment insurance contribution, sales tax rate, targeted business incentives, apportionment factors, and throwback rules.
- For manufacturing, the paper can also construct effective tax rates using apportionment rules and firm-specific factor weights.

Interpretation: Controlling for other state tax instruments is important because corporate and personal taxes are not the only fiscal policies that might affect firm location.

3.4 Outcome variables

- **Extensive margin:** number of establishments a firm has in a state-year.
- **Intensive margin:** log employment at the establishment-year level.
- **Capital margin:** log capital stock in manufacturing establishments.
- **Reallocation:** responses to own-state taxes versus average taxes in other states where the firm has operations.

Interpretation: The paper’s data allow the authors to distinguish whether taxes affect where establishments exist, how much activity occurs inside establishments, and whether activity is moved across states.

4 Empirical specifications

4.1 Baseline extensive-margin specification

At the firm-state-year level, the core extensive-margin regression can be written as

$$y_{fst} = \beta_1 \tau_{st}^C \times CCorp_f + \beta_2 \tau_{st}^C \times Pass_f + \beta_3 \tau_{st}^P \times CCorp_f + \beta_4 \tau_{st}^P \times Pass_f + X'_{st} \Gamma + \alpha_{fs} + \lambda_t + \varepsilon_{fst}. \quad (1)$$

- y_{fst} is the number of establishments firm f has in state s in year t .
- $CCorp_f$ and $Pass_f$ are indicators for the firm’s tax status.
- α_{fs} are firm-state fixed effects.
- λ_t are year fixed effects.
- X_{st} contains state-year tax and policy controls.

Interpretation: The key coefficients are β_1 for C corporations responding to corporate taxes and β_4 for pass-through firms responding to personal taxes.

4.2 Baseline intensive-margin specification

At the establishment-year level, the intensive-margin regression is similar:

$$y_{est} = \beta_1 \tau_{st}^C \times CCorp_f + \beta_2 \tau_{st}^C \times Pass_f + \beta_3 \tau_{st}^P \times CCorp_f + \beta_4 \tau_{st}^P \times Pass_f + X'_{st} \Gamma + \alpha_e + \lambda_t + \varepsilon_{est}, \quad (2)$$

where y_{est} is log employment or log capital for establishment e .

Interpretation: Establishment fixed effects compare the same establishment over time as state taxes change.

4.3 Reallocation across states

To test whether firms shift activity to other states, the paper includes average tax rates in other states where the firm operates:

$$y_{fst} = \cdots + \delta_1 \bar{\tau}_{f,-s,t}^C \times CCorp_f + \delta_2 \bar{\tau}_{f,-s,t}^P \times Pass_f + \varepsilon_{fst}. \quad (3)$$

- $\bar{\tau}_{f,-s,t}$ is the employment-weighted average tax rate in the firm's other states.
- A positive coefficient on other-state tax rates implies substitution toward state s when other states become more expensive.

Interpretation: This specification directly measures reallocation. If a firm's other states raise taxes, the firm should increase activity in the focal state.

4.4 Large tax changes and dynamic responses

The paper also studies large tax changes using event-time specifications:

$$y_{it} = \sum_{k=-2}^2 \beta_k \mathbf{1}\{t - T_s = k\} \times Treatment_s + FE + Controls + \varepsilon_{it}. \quad (4)$$

Interpretation: These specifications show whether activity changes before or after large tax increases and whether responses persist over time.

5 Identification and research design

5.1 Within-firm and within-establishment variation

- Firm-state fixed effects absorb permanent differences in the attractiveness of a state for a given firm.
- Establishment fixed effects absorb time-invariant establishment productivity, industry, and location characteristics.
- Year fixed effects absorb national macroeconomic shocks.
- The remaining variation is within-firm or within-establishment changes in state tax rates.

Interpretation: The design is stronger than a cross-state aggregate regression because it compares the same firm or establishment before and after tax changes.

5.2 Tax-code placebo structure

- Corporate tax rates should matter for C corporations.
- Corporate tax rates should not matter for pass-through entities.
- Personal tax rates should matter for pass-through entities.
- Personal tax rates should not matter for C corporations.

Interpretation: The empirical results largely match this cross-tax-code prediction, which helps address concerns that tax changes proxy for general economic conditions.

5.3 Reallocation versus contraction

- Own-state tax increases reduce activity in that state.
- Other-state tax increases raise activity in the focal state.
- The paper concludes that reallocation accounts for around half of the effect.

Interpretation: This is economically important because the incidence of state taxation depends on whether firms reduce total activity or simply shift activity across state borders.

5.4 Limitations

- State tax changes are not randomly assigned.
- The sample focuses on large multistate firms, which may be more mobile than the average business.
- Responses of smaller and single-state establishments are harder to identify.
- The analysis measures relatively short-run responses; very long-run firm entry and organizational-form changes may differ.

Interpretation: The paper is best interpreted as evidence about large multistate firms' real activity responses to state taxation.

6 Tables and figures

6.1 Figure 1 and Table 1: sample composition and summary statistics

Read:

- Figure 1 shows the decline of C corporations and rise of pass-through entities from 1977 to 2011.
- In the sample of large multistate firms, C corporations remain the dominant form, but pass-through entities become increasingly important.
- Table 1 reports the scale of the sample: about 27.6 million establishment-year observations in the LBD and 854,700 manufacturing establishment-year observations in ASM/CMF.
- C corporations are larger on average than pass-through entities, especially in the right tail of establishments and employees.

Economic message: The sample contains large multistate firms, with enough variation across legal forms to separately test corporate and personal tax effects.

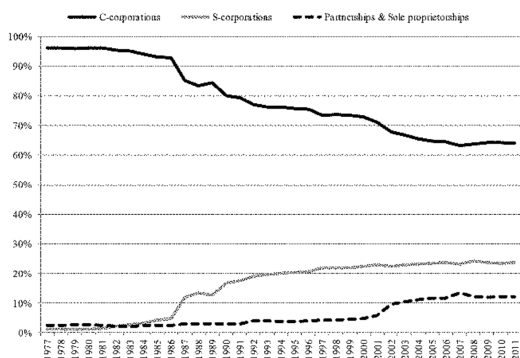


FIG. 1.—Legal forms of organization over time. This figure plots the percentage of companies whose legal form of organization is C corporation, S corporation, and partnership or sole proprietorship. The sample includes all multiunit companies in the Longitudinal Business Database (LBD) with at least 100 employees and establishments in at least two states. The sample period is from 1977 until 2011.

Figure 1: legal forms over time

	LBD (All Sectors)			ASM/CMF (Manufacturing)		
	All	C-Corp	Pass-Through	All	C-Corp	Pass-Through
A. Establishment Level						
# employees:						
Mean	49	50	36	311	318	153
Median	11	11	11	142	145	92
Standard deviation	228	235	112	757	772	209
Capital stock (\$1997):						
Mean				42,586	43,946	12,850
Median				11,141	11,495	5,886
Standard deviation				170,941	174,555	31,723
Observations	27,600,100	25,271,400	2,328,700	854,700	817,300	37,400
B. Firm-State Level						
# establishments:						
Mean	6.56	7.06	3.72	1.76	1.79	1.27
Median	1.25	1.26	1.21	1.00	1.00	1.00
Standard deviation	25.67	27.37	11.54	1.94	1.99	.88
# employees:						
Mean	320	352	133	546	569	195
Median	58	61	46	201	210	107
Standard deviation	1550	1,669	411	1,970	2,029	318
Capital stock (\$1997):						
Mean				74,765	78,516	16,354
Median				15,805	16,785	6,691
Standard deviation				362,123	373,066	47,513
Observations	4,207,200	3,580,600	626,600	486,800	457,400	29,400
Observations (including zeros)	32,997,200	25,225,300	7,771,900	5,325,600	4,758,400	567,200
C. Firm Level						
# states:						
Mean	6.50	7.24	4.11	4.66	4.90	2.64
Median	3.37	3.70	2.30	2.28	2.29	2.00
Standard deviation	8.3	9.06	4.31	4.96	5.17	1.44
# establishments:						
Mean	42.66	51.09	15.28	8.18	8.76	3.36
Median	8.04	8.41	6.25	3.36	3.70	2.32
Standard deviation	239.83	269.31	88.43	15.02	15.76	2.67
# employees:						
Mean	2,075	2,547	546	2,547	2,790	516
Median	359	417	252	683	767	295
Standard deviation	11,841	13,480	1,583	9,251	9,756	715
Capital stock (\$1997):						
Mean				348,551	384,950	43,204
Median				49,606	57,151	18,885
Standard deviation				1,513,534	1,596,842	112,904
Observations	647,000	494,600	152,400	104,400	93,300	11,100

NOTE.—In panel A, observations are at the establishment-year level. All refers to all establishments; C-Corp refers to establishments belonging to C corporations; Pass-Through refers to establishments belonging to pass-through entities (S corporations, partnerships, and sole proprietorships). LBD refers to establishments in the Longitudinal Business Database; ASM/CMF refers to establishments in the Annual Survey of Manufactures and the Census of Manufactures. Capital stock is constructed using the perpetual inventory method. In panels B and C, observations are aggregated into the firm-state-year and firm-year level.

Table 1: summary statistics

6.2 Table 2 and Figure 2: tax variables and tax-rate variation

Read:

- Table 2 shows substantial cross-state and time-series variation in corporate and personal tax rates.
- Corporate tax rates average 6.85 percent; personal tax rates average 5.29 percent.
- Figure 2 shows that corporate and personal income tax rates evolve differently over time.

- Personal tax rates are more compressed over time, while corporate rates fluctuate differently across states.

Economic message: Independent variation in corporate and personal taxes is essential for identifying differential effects by organizational form.

TABLE 2
TAX VARIABLES

	Mean	Standard Deviation	25th Percentile	50th Percentile	75th Percentile
τ_c	6.85	2.95	6.00	7.00	8.90
τ_p	5.29	3.34	3.02	5.82	7.50
Property tax share	.12	.05	.09	.12	.15
UI contribution	699	433	378	585	855
Log(UI contribution)	6.38	.58	5.93	6.37	6.75
Sales tax rate	4.44	1.82	4.00	5.00	6.00
Tax incentives index	21.13	6.42	16.00	23.00	26.00
Payroll apportionment factor	26.82	9.95	25.00	33.33	33.33
Property apportionment factor	26.82	9.95	25.00	33.33	33.33
Sales apportionment factor	46.35	19.88	33.34	33.34	50.00
Throwback rule	.60	.49	.00	1.00	1.00

NOTE.—This table shows summary statistics for the tax variables used in the analysis. τ_c is the top corporate income tax rate (in percent). τ_p is the top personal income tax rate (in percent). Property tax share is the ratio of the total amount of property taxes (collected by the state and local governments) divided by total revenues (of the state and local governments). UI contribution is the top unemployment insurance (UI) rate multiplied by the maximum base wage (in percent). Tax incentives index is an index of tax incentives that adds one index point for each of the 33 tax incentives compiled in the *Site Selection* magazine. Sales tax rate is the sales tax rate in percentage points. Payroll apportionment factor is the apportionment percentage attributed to payroll in percentage points. Property apportionment factor and sales apportionment factor are defined similarly with respect to property and sales, respectively. Throwback is an indicator variable equal to one if the state has a throwback (or a throwout) rule. Summary statistics are computed using all available state-year observations from 1977 to 2011.

Table 2: tax variables

6.3 Figure 3: tax-rate changes

Read:

- Figure 3 plots the number of increases and decreases in corporate and personal income tax rates by year.
- Corporate tax changes are concentrated in some early sample years, while personal tax changes are more frequent in later periods.
- The figure confirms that the identification uses many state-year tax changes rather than one isolated reform.

Economic message: The staggered timing of state tax changes provides the variation needed for fixed-effect estimates.

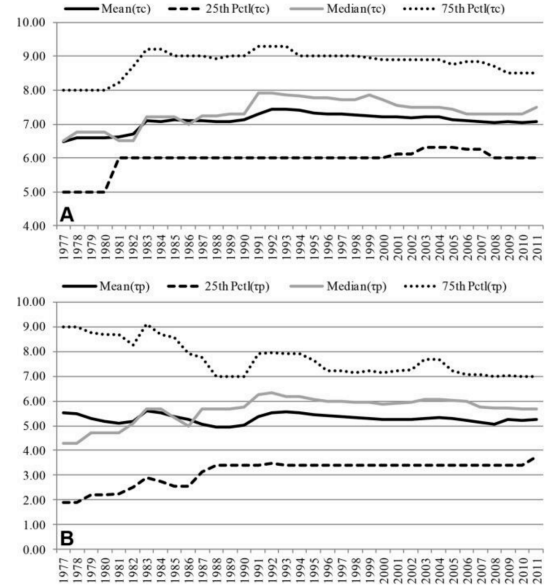


FIG. 2.—Corporate and personal income tax rates over time. This figure plots the evolution of the mean and quartiles of the corporate income tax rate (τ_c , panel A) and personal income tax rate (τ_p , panel B), respectively, across all states from 1977 to 2011.

Figure 2: corporate and personal tax rates

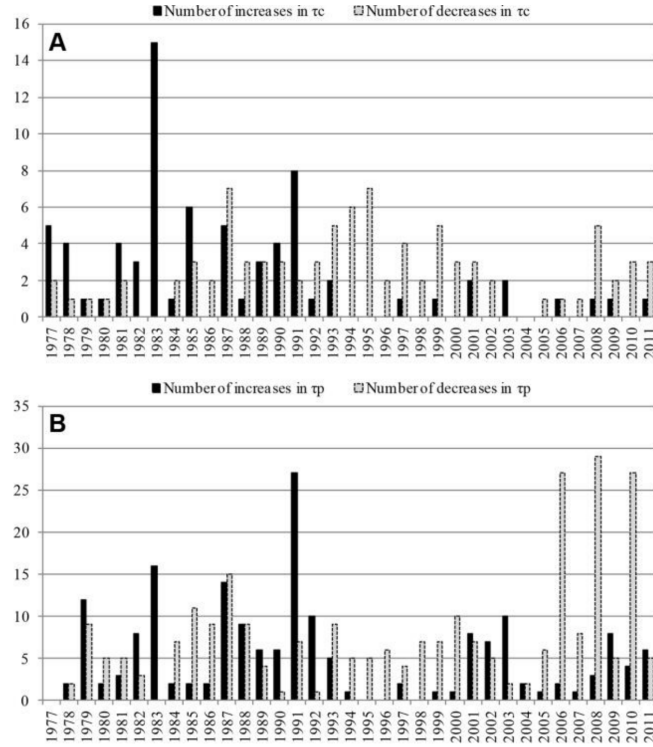


FIG. 3.—Changes in corporate and personal income tax rates over time. This figure plots the number of changes in the corporate income tax rate (τ_c , panel A) and personal income tax rate (τ_p , panel B), respectively, across all states from 1977 to 2011.

Figure 3: tax-rate changes over time

6.4 Table 3: main results

Read:

- Table 3 is the main results table.
- On the extensive margin, C corporation establishments fall when corporate tax rates rise.
- Pass-through establishments respond to personal tax rates, not corporate tax rates.
- On the intensive margin, log employment and manufacturing capital exhibit the same organizational-form-specific pattern.
- Coefficients on sales taxes, UI contributions, property tax share, and incentives are included as controls.

Economic message: The core pattern is a clean cross-tax-code falsification: each organizational form responds to the tax code that applies to it.

	EXTENSIVE MARGIN			INTENSIVE MARGIN			
	# Establishments	# Establishments	# Establishments	All Sectors		Manufacturing	
	(1)	(2)	(3)	Log(Employees)	Log(Employees)	Log(Employees)	Log(Capital)
$\tau_c \times \text{C-Corp}$	-.037*** (.003)	-.031*** (.003)	-.023*** (.002)	-.0041*** (.0005)	-.0038*** (.0005)	-.0035*** (.0011)	-.0024*** (.0008)
$\tau_c \times \text{Pass-Through}$	-.002 (.003)	-.000 (.003)	-.002 (.002)	-.0004 (.0010)	-.0001 (.0010)	-.0003 (.0023)	.0000 (.0015)
$\tau_p \times \text{C-Corp}$	-.003 (.002)	-.001 (.002)	-.002 (.002)	-.0007 (.0004)	-.0003 (.0004)	-.0010 (.0008)	-.0002 (.0005)
$\tau_p \times \text{Pass-Through}$	-.016*** (.003)	-.010*** (.003)	-.010*** (.002)	-.0024*** (.0009)	-.0019*** (.0009)	-.0026 (.0022)	-.0015 (.0015)
Sales tax rate	-.001 (.005)	-.000 (.005)	.000 (.005)	-.0005 (.0007)	-.0005 (.0007)	.0000 (.0016)	-.0004 (.0011)
Log(UI contribution)	-.189*** (.008)	-.183*** (.008)	-.118*** (.006)	-.0223*** (.0009)	-.0110*** (.0014)	-.0089*** (.0040)	-.0063*** (.0026)
Property tax share	-.386*** (.018)	-.371*** (.018)	-.175*** (.026)	-.0128 (.0107)	-.0032 (.0107)	-.0366 (.0248)	-.0245 (.0165)

Table 3: main results, part 1

Tax incentives index	.002* (.001)	.002** (.001)	.002** (.001)	.0008*** (.0001)	.0009*** (.0001)	.0072*** (.0004)	.0018*** (.0002)
G-Corp	.348*** (.015)	.352*** (.015)	.290*** (.010)	.0015 (.0051)	.0028 (.0051)	.0010 (.0163)	-.0080 (.0105)
Log(GDP)		.225*** (.025)			.241*** (.0054)		
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm-state fixed effects	Yes	Yes	Yes	No	No	No	No
Establishment fixed effects	No	No	No	Yes	Yes	Yes	Yes
Regression type	OLS	OLS	Poisson	OLS	OLS	OLS	OLS
R ²	.73	.73	. . .	.88	.88	.92	.96
Observations	32,997,200	32,997,200	32,997,200	27,600,100	27,600,100	854,700	854,700

NOTE.—In cols. 1–3, the dependent variable is the number of establishments of a given firm in a given state and year. The number of establishments is set to zero if an active company has no operation in the state. In cols. 4–6, the dependent variable is the logarithm of the number of employees at the establishment. In col. 7, the dependent variable is the logarithm of the establishment's capital stock. Capital stock is constructed using the perpetual inventory method. G-Corp is a dummy variable that equals one if a company is a C corporation, and Pass-Through is a dummy variable that equals one if a company is an S corporation, partnership, or sole proprietorship for tax purposes. GDP is the state's gross domestic product (from the BEA). The other variables are defined in table 2. All regressions include dummy variables for Michigan, Ohio (post-2005), Texas (post-1991), and Washington, interacted with G-Corp and Pass-Through. In cols. 6–7, the sample is restricted to establishments in the ASM/CMF. The sample period is 1977–2011. Standard errors are clustered at the state level.

* Significant at the 10 percent level.

** Significant at the 5 percent level.

*** Significant at the 1 percent level.

Table 3: main results, part 2

6.5 Table 4 and Table 5: reallocation and large-tax-change design

Read:

- Table 4 shows that own-state tax increases reduce activity, while tax increases in other states raise activity in the focal state.
- This implies that firms reallocate activity across states in response to relative tax changes.
- Table 5 uses large corporate and personal tax changes in a difference-in-differences framework.
- The dynamic large-tax-change evidence supports the fixed-effect regression results.

Economic message: State taxes move the geography of firm activity. The effect is not simply a firm-level shrinkage effect.

	# ESTABLISHMENTS	LOG(EMPLOYEES)	
	(1)	(2)	(3)
$\tau_c \times \text{C-Corp}$	-.037*** (.003)	-.025*** (.002)	-.0044*** (.0005)
$\tau_c \times \text{Pass-Through}$	-.002 (.003)	-.002 (.002)	-.0000 (.0011)
$\tau_p \times \text{C-Corp}$	-.003 (.002)	-.002 (.002)	-.0004 (.0004)
$\tau_p \times \text{Pass-Through}$	-.016*** (.003)	-.012*** (.002)	-.0024** (.0010)
$\tilde{\tau}_c$ (other states) $\times$ C-Corp	.018*** (.002)	.010*** (.003)	.0021*** (.0006)
$\tilde{\tau}_c$ (other states) $\times$ Pass-Through	-.000 (.002)	.001 (.002)	-.0001 (.0013)
$\tilde{\tau}_p$ (other states) $\times$ C-Corp	.001 (.002)	.001 (.002)	.0000 (.0006)
$\tilde{\tau}_p$ (other states) $\times$ Pass-Through	.006*** (.002)	.005* (.003)	.0011 (.0014)
Controls	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Firm-state fixed effects	Yes	Yes	No
Establishment fixed effects	No	No	Yes
Regression type	OLS	Poisson	OLS
R ²	.73	. . .	.88
Observations	32,997,200	32,997,200	27,600,100

NOTE.—This table presents variants of the regressions in cols. 1, 3, and 4 of table 3. $\tilde{\tau}_c$ (other states) refers to the average corporate income tax rate in all other states in which the company has operations. The average is computed using the share of the company's employees in each state as weights. $\tilde{\tau}_p$ (other states) is computed analogously with respect to the personal income tax rate. The sample period is 1977–2011. Standard errors are clustered at the state level.

* Significant at the 10 percent level.

** Significant at the 5 percent level.

*** Significant at the 1 percent level.

ditional details and robustness tests on our analysis may be found there.

Table 4: reallocation across states

	# ESTABLISHMENTS							
	Large Cuts in τ_c		Large Increases in τ_c		Large Cuts in τ_p		Large Increases in τ_p	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Treatment	.027*** (.005)		-.014*** (.003)		.018*** (.004)		-.005 (.003)	
Treatment (-2)		-.002 (.004)		.000 (.003)		.005 (.003)		.001 (.003)
Treatment (-1)		-.002 (.006)		.004 (.003)		.005 (.003)		.003 (.004)
Treatment (0)		.017** (.007)		-.007* (.004)		.015*** (.004)		-.002 (.005)
Treatment (+1)		.031*** (.008)		-.017*** (.004)		.028*** (.005)		.003 (.005)
Treatment (+2)		.030*** (.009)		-.017*** (.005)		.026*** (.007)		-.005 (.005)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm-state fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R ²	.88	.88	.92	.92	.87	.87	.86	.86
Observations	1,748,600	1,748,600	5,144,600	5,144,600	3,561,900	3,561,900	4,697,400	4,697,400

NOTE.—This table estimates the treatment effect of large tax changes (increases or decreases in tax rates that are at least 100 basis points) on the number of establishments. In cols. 1 and 2, the treatments are large decreases in the corporate income tax rate. The sample includes all firm-state-year observations in the treated states 3 years before and after the treatments (i.e., the treatment group consists of C corporations; the control group consists of pass-through entities). In col. 1, Treatment is a dummy variable equal to one for C corporations in the years following the treatment. In col. 2, Treatment (-2) is a dummy variable equal to one for C corporations 2 years prior to the treatment. Treatment (-1), Treatment (0), Treatment (+1), and Treatment (+2) are defined similarly. The analysis in cols. 3–8 is conducted analogously with respect to large increases in the corporate income tax rate and large decreases/increases in the personal income tax rate, respectively. The sample period is 1977–2011. Standard errors are clustered at the state level.

* Significant at the 10 percent level.

** Significant at the 5 percent level.

*** Significant at the 1 percent level.

Table 5: large-tax-change DiD, extensive margin

6.6 Figure 4 and Figure 6: dynamic responses

Read:

- Figure 4 plots dynamic extensive-margin responses around large tax changes.

- The response is visible after large tax changes rather than before them.
- Figure 6 extends the dynamic horizon and shows cumulative effects of tax changes.
- C corporation activity declines following corporate tax increases; pass-through activity declines following personal tax increases.

Economic message: Dynamic figures are important because they show that the response is not driven by pre-existing trends immediately before the tax changes.

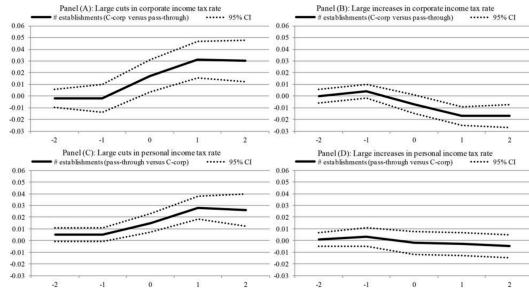
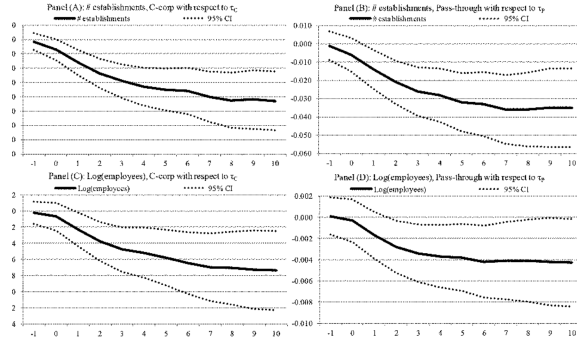


FIG. 4.—Extensive margin: dynamic effect of large tax changes. This figure plots the coefficients (and 95 percent confidence intervals) corresponding to the dynamic analysis provided in table 5. See the note to table 5 for details.

Figure 4: dynamic extensive-margin effects



10-year dynamics. Panel A plots the cumulative coefficients (and 95 percent confidence intervals) of $\tau_c \times$ C-Corp in an extended version of ϵ regression at the extensive margin, in which we include 10 lags and one lead of $\tau_c \times$ C-Corp. Panel B is analogous with respect to $\tau_p \times$ Pass-through. Panels C and D are analogous with respect to the intensive margin regression.

Figure 6: cumulative dynamic effects

6.7 Tables 7–9: large-tax-change intensive-margin results

Read:

- Tables 7–9 show intensive-margin versions of the large-tax-change analysis.
- The results show changes in log employment, and in manufacturing analyses also capital.
- The pattern again maps corporate taxes to C corporations and personal taxes to pass-through entities.
- These tables help confirm that the response is not limited to establishment entry and exit.

Economic message: Taxes affect both whether firms operate in a state and how much activity they conduct inside existing establishments.

	# ESTABLISHMENTS					
	Large Cuts in τ_c (1)	Large Increases in τ_c (2)	Large Cuts in τ_p (3)	Large Increases in τ_p (4)	Large Cuts in τ_p (5)	Large Increases in τ_p (6)
Treatment (exogenous)	.029*** (.007)	-.016*** (.005)		.019*** (.005)		-.005 (.005)
Treatment (ERTAS1)			-.014** (.007)			
Treatment (TRAS6)				.019*** (.007)		-.007 (.017)
Treatment (other exogenous)			-.017*** (.005)	.019** (.008)		-.004 (.005)
Treatment (endogenous)	.033** (.013)	-.015** (.007)	.018* (.007)	.018* (.010)	-.004 (.015)	-.004 (.015)
Treatment (unclassified)	.022** (.011)	-.014*** (.005)	.016*** (.003)	.016*** (.005)	-.003 (.006)	-.003 (.006)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Firm-state fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
N	88	92	87	87	86	86
Observations	1,748,600	3,144,600	3,561,900	3,561,900	4,697,400	4,697,400

NOTE.—This table presents variants of the regressions in cols. 1, 3, 5, and 7 of table 5, decomposing the treatment into exogenous, endogenous, and other types of treatments using the methodology of Romer and Romer (2010). ERTAS1 refers to the Economic Recovery Tax Act of 1981; TRAS6 refers to the Tax Reform Act of 1986. The sample period is 1977–2011. Standard errors are clustered at the state level.

* Significant at the 10 percent level.

** Significant at the 5 percent level.

*** Significant at the 1 percent level.

Table 7: additional extensive-margin DiD

	LOG(EMPLOYEES)					
	Large Cuts in τ_c (1)	Large Increases in τ_c (2)	Large Cuts in τ_p (3)	Large Increases in τ_p (4)	Large Cuts in τ_p (5)	Large Increases in τ_p (6)
Treatment	.0032*** (.0007)	-.0034*** (.0006)	.0009 (.0006)	-.0002 (.0007)	-.0022*** (.0004)	.0003 (.0004)
Treatment (-2)		-.0007 (.0010)	.0007 (.0007)	.0002 (.0007)		.0003 (.0006)
Treatment (-1)		-.0016 (.0011)	.0011 (.0007)	-.0003 (.0008)		.0002 (.0008)
Treatment (0)		.0015 (.0011)	-.0018** (.0008)	.0007 (.0009)		-.0011 (.0007)
Treatment (+1)		.0027** (.0012)	-.0045*** (.0009)	.0008 (.0010)		-.0027*** (.0007)
Treatment (+2)		.0033*** (.0011)	-.0051*** (.0011)	.0014 (.0011)		-.0033*** (.0008)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Establishment fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
N	94	94	95	95	94	94
Observations	1,926,800	1,926,800	1,950,600	1,950,600	2,420,100	2,420,100

NOTE.—This table presents variants of the regressions in table 5, except that the analysis is conducted at the intensive margin, that is, at the establishment-year level. The dependent variable is the logarithm of the number of employees at the establishment. The sample period is 1977–2011. Standard errors are clustered at the state level.

* Significant at the 10 percent level.

** Significant at the 5 percent level.

*** Significant at the 1 percent level.

Table 8: intensive-margin DiD

	LOG(EMPLOYEES)					
	Large Cuts in τ_c (1)	Large Increases in τ_c (2)		Large Cuts in τ_p (4)	Large Increases in τ_p (6)	
Treatment (exogenous)	.0030*** (.0009)	-.0036*** (.0006)		.0009 (.0008)	-.0022*** (.0005)	
Treatment (ERTA81)			-.0043*** (.0016)			
Treatment (TRA86)				.0010 (.0012)		-.0018 (.0019)
Treatment (other exogenous)			-.0035*** (.0007)	.0009 (.0008)		-.0025*** (.0006)
Treatment (endogenous)	.0037* (.0022)	-.0029*** (.0008)	-.0028*** (.0008)	.0008 (.0009)	-.0018* (.0010)	-.0019* (.0010)
Treatment (unclassified)	.0033* (.0017)	-.0031** (.0013)	-.0032** (.0013)	.0009 (.0013)	-.0021** (.0010)	-.0021** (.0010)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Establishment fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
R^2	.94	.95	.95	.93	.94	.94
Observations	1,326,800	1,950,600	1,950,600	2,420,100	3,364,500	3,364,500

NOTE.—This table presents regressions similar to those in table 8, except that the analysis is conducted at the intensive margin, that is, at the establishment-clustered at the state level.

* Significant at the 10 percent level.

** Significant at the 5 percent level.

*** Significant at the 1 percent level.

Table 9: intensive-margin DiD, alternative treatment definitions

6.8 Tables 13–15: heterogeneity, public companies, and general equilibrium

Read:

- Table 13 shows stronger effects in footloose, tradable, and labor-intensive industries.
- Table 14 shows that public companies respond to taxes, including differential patterns for multinational firms.
- Table 15 aggregates employment at the state-legal-form-year level and asks whether other establishments absorb the displaced activity.
- The table does not show offsetting employment increases among smaller and single-state establishments.

Economic message: The heterogeneity results support the reallocation interpretation. Firms in industries that can relocate production or serve customers from elsewhere respond most strongly.

	# Establishments	Log(Employees)
$\tau_c \times C\text{-Corp}$	-.013** (.005)	-.0014 (.0012)
$\tau_c \times C\text{-Corp} \times \text{Footloose industry}$	-.031*** (.011)	-.0040*** (.0015)
$\tau_c \times C\text{-Corp} \times \text{Tradable industry}$	-.089*** (.035)	-.0082*** (.0035)
$\tau_c \times C\text{-Corp} \times \text{Labor-intensive industry}$	-.042*** (.013)	-.0051*** (.0009)
$\tau_c \times \text{Pass-Through}$	.001 (.001)	-.0001 (.0002)
$\tau_c \times \text{Pass-Through} \times \text{Footloose industry}$	-.000 (.010)	-.0003 (.0010)
$\tau_c \times \text{Pass-Through} \times \text{Tradable industry}$	-.005 (.010)	-.0007 (.0010)
$\tau_c \times \text{Pass-Through} \times \text{Labor-intensive industry}$	-.004 (.010)	-.0005 (.0006)
$\tau_p \times C\text{-Corp}$	.000 (.001)	-.0005 (.0020)
$\tau_p \times C\text{-Corp} \times \text{Footloose industry}$	-.002 (.009)	-.0005 (.0010)
$\tau_p \times C\text{-Corp} \times \text{Tradable industry}$	-.006 (.010)	-.0009 (.0007)
$\tau_p \times C\text{-Corp} \times \text{Labor-intensive industry}$	-.010 (.007)	-.0005 (.0005)
$\tau_p \times \text{Pass-Through}$	-.005** (.002)	-.0008 (.0005)
$\tau_p \times \text{Pass-Through} \times \text{Footloose industry}$	-.014** (.006)	-.0026** (.0012)
$\tau_p \times \text{Pass-Through} \times \text{Tradable industry}$	-.023* (.012)	-.0050** (.0021)
$\tau_p \times \text{Pass-Through} \times \text{Labor-intensive industry}$	-.018*** (.006)	-.0024** (.0011)
Controls	Yes	Yes
Year fixed effects	Yes	Yes
Firm-state fixed effects	Yes	No
Establishment fixed effects	No	Yes
R^2	.73	.88
Observations	39 947 900	97 600 100

Table 13: cross-sectional heterogeneity

	# ESTABLISHMENTS		LOG(EMPLOYEES)	
	(1)	(2)	(3)	(4)
$\tau_c \times C\text{-Corp}$	-.044*** (.009)		-.0048*** (.0016)	
$\tau_p \times C\text{-Corp}$	-.003 (.004)		-.0005 (.0012)	
$\tau_c \times C\text{-Corp} \times \text{Domestic}$		-.035*** (.010)		-.0040*** (.0016)
$\tau_c \times C\text{-Corp} \times \text{Multinational}$		-.056*** (.011)		-.0052*** (.0016)
$\tau_p \times C\text{-Corp} \times \text{Domestic}$		-.003 (.004)		-.0005 (.0012)
$\tau_p \times C\text{-Corp} \times \text{Multinational}$		-.002 (.004)		-.0004 (.0012)
Multinational		.245*** (.031)		.0211*** (.0020)
Controls	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Firm-state fixed effects	Yes	Yes	No	No
Establishment fixed effects	No	No	Yes	Yes
R^2	.83	.83	.92	.92
Observations	3,370,600	3,370,600	8,428,900	8,428,900

NOTE.—This table presents variants of the regressions in cols. 1 and 4 of table 3, restricting the sample to public companies. Public companies are those with coverage in Standard & Poor's Compustat. Compustat is matched to the LBD using the SSEL-Compustat Bridge maintained by the US Census Bureau. Multinational is a dummy variable equal to one if the company has nondomestic segments in the Compustat Segment file. The sample period is 1977–2011. Standard errors are clustered at the state level.

* Significant at the 10 percent level.

** Significant at the 5 percent level.

Table 14: public companies

TABLE 15
GENERAL EQUILIBRIUM

	LOG(EMPLOYEES)	
	Establishments of Multistate Firms with More than 100 Employees (1)	Other Establishments (2)
$\tau_C \times \text{C-Corp}$	-.0039** (.0016)	-.0014 (.0010)
$\tau_C \times \text{Pass-Through}$	.0006 (.0015)	.0003 (.0011)
$\tau_P \times \text{C-Corp}$	-.0003 (.0009)	.0003 (.0006)
$\tau_P \times \text{Pass-Through}$	-.0018** (.0008)	-.0006 (.0006)
Controls	Yes	Yes
Year fixed effects	Yes	Yes
LFO-state fixed effects	Yes	Yes
R^2	.92	.89
Observations	3,600	3,600

NOTE.—This table presents state-level analogues of the regressions in table 3. The unit of observation is the state-LFO-year. Employment is aggregated at the state-LFO-year level using all establishments in our sample (col. 1) and all other LBD establishments (col. 2). The sample period is 1977–2011. Standard errors are clustered at the state level.

* Significant at the 10 percent level.
** Significant at the 5 percent level.

Table 15: general equilibrium evidence

7 Detailed result synthesis

7.1 Extensive-margin response

The number of establishments responds strongly to the relevant state tax rate. C corporations reduce establishment counts in high corporate-tax states. Pass-through entities reduce establishment counts in high personal-tax states. The cross-response coefficients are small, which reinforces the tax-code interpretation.

7.2 Intensive-margin response

Employment and manufacturing capital respond on the intensive margin. This shows that taxes do not only affect discrete entry and exit decisions; they also change the scale of operations inside existing establishments.

7.3 Reallocation margin

A central quantitative result is that around half of the response reflects reallocation to other states. This means state tax policy affects the geographic allocation of employment and capital across states, not just total firm scale.

7.4 Industry heterogeneity

Footloose and tradable industries are more sensitive to taxes. This supports the mechanism that firms respond when they have technologically or commercially feasible alternatives to produce elsewhere.

8 Short summary

- Giroud and Rauh study state taxation using Census microdata on multistate firms.
- They exploit the distinction between C corporations and pass-through entities.
- C corporations respond to corporate tax rates, not personal tax rates.
- Pass-through entities respond to personal tax rates, not corporate tax rates.
- Outcomes include establishments, employment, and manufacturing capital.
- Responses occur on both extensive and intensive margins.

- About half of the effect reflects reallocation across states within firms.
- Responses are strongest in tradable and footloose industries.
- The paper provides micro evidence that state taxes affect real business activity and geographic allocation.

9 Conclusion

9.1 Core conclusion

State tax policy has economically meaningful effects on business activity. The effects are visible in employment, establishments, and capital, and they line up with the tax code that applies to each organizational form.

9.2 Economic intuition

Large multistate firms can move production and employment across state lines. When a state raises the tax rate that applies to a firm's organizational form, that state becomes less attractive for activity. Firms respond by reducing activity there and increasing activity elsewhere.

9.3 Policy takeaway

State governments face a real tax-base mobility constraint. Higher state business taxes can reduce local employment and capital, especially in industries where production is tradable or footloose. The paper's evidence shows that the relevant margin is not only entry and exit of firms but the internal reallocation of activity by multistate firms.